Assignment: 4 Solution

Q.1. Which of the following properties holds true for both classical sets and fuzzy sets?

(A) Law of contradiction

(B) Law of excluded middle

(C) Absorption of complement

(D) Distributivity

(E) None of the above

Solution: (D)

Q.2. In the context of DeMorgan's law, what is the result of taking the complement of the union of two fuzzy sets A and B?

(A) Intersection of A complement and B complement

(B) Union of A complement and B complement

(C) Intersection of A and B

(D) Union of A and B

(E) None of the above

Solution: (A)

Q.3. In the context of DeMorgan's lawof intersection states that the complement of the intersection of two sets is equal to what?

(A) Union of the complements of the sets

(B) Intersection of the complements of the sets

(C) Complement of the union of the sets

(D) Complement of one of the sets

(E) None of the above

Solution: (A)

Q.4. What does δ represent in the context of computing the distance between fuzzy sets A and B?

(A) The universe of discourse for fuzzy sets A and B

(B) The maximum membership value between fuzzy sets A and B

(C) The difference between the corresponding generic variables of fuzzy sets A and B

(D) None of the above

Solution: (C)

Q.5. How is the membership value for a specific delta calculated in the table for finding the distance between two fuzzy sets?

(A) By averaging the membership values for all pairs with the same delta

(B) By taking the maximum of the membership values for all pairs with the same delta

(C) By summing the membership values for all pairs with the same delta

(D) None of the above

Solution: (B)

Q.6. What conclusion is drawn from the comparison of the distances between fuzzy sets A and B and between fuzzy sets B and A?

(A) The distances are equal

(B) The distances are never equal

(C) The distances cannot be calculated accurately

(D) None of the above

Solution: (A)

Q.7. What is the purpose of computing the distance between two fuzzy sets?

(A) To find the union of the sets

(B) To measure the similarity between the sets

(C) To identify the intersection of the sets

(D) To determine the complement of the sets

(E) None of the above

Solution: (B)

Q.8. What is a fuzzy number?

(A) A fuzzy set that satisfies normality and convexity conditions

(B) A discrete set with no variability

(C) A singleton fuzzy set with infinite elements

(D) None of the above

Solution: (A)

Q.9. How is a fuzzy number different from a crisp number?

(A) Crisp numbers have infinite membership values, while fuzzy numbers have only one membership value

(B) Crisp numbers have variability, while fuzzy numbers have fixed membership values

(C) Crisp numbers are singletons with only one membership value, while fuzzy numbers are sets with variable membership values

(D) None of the above

Solution: (C)

Q.10. How are arithmetic operations, such as addition, performed on fuzzy numbers?

(A) By finding the minimum of corresponding membership values and then taking the maximum of the results

(B) By finding the maximum of corresponding membership values and then taking the minimum of the results

(C) By finding the sum of corresponding membership values

(D) By finding the product of corresponding membership values

(E) None of the above

Solution: (A)